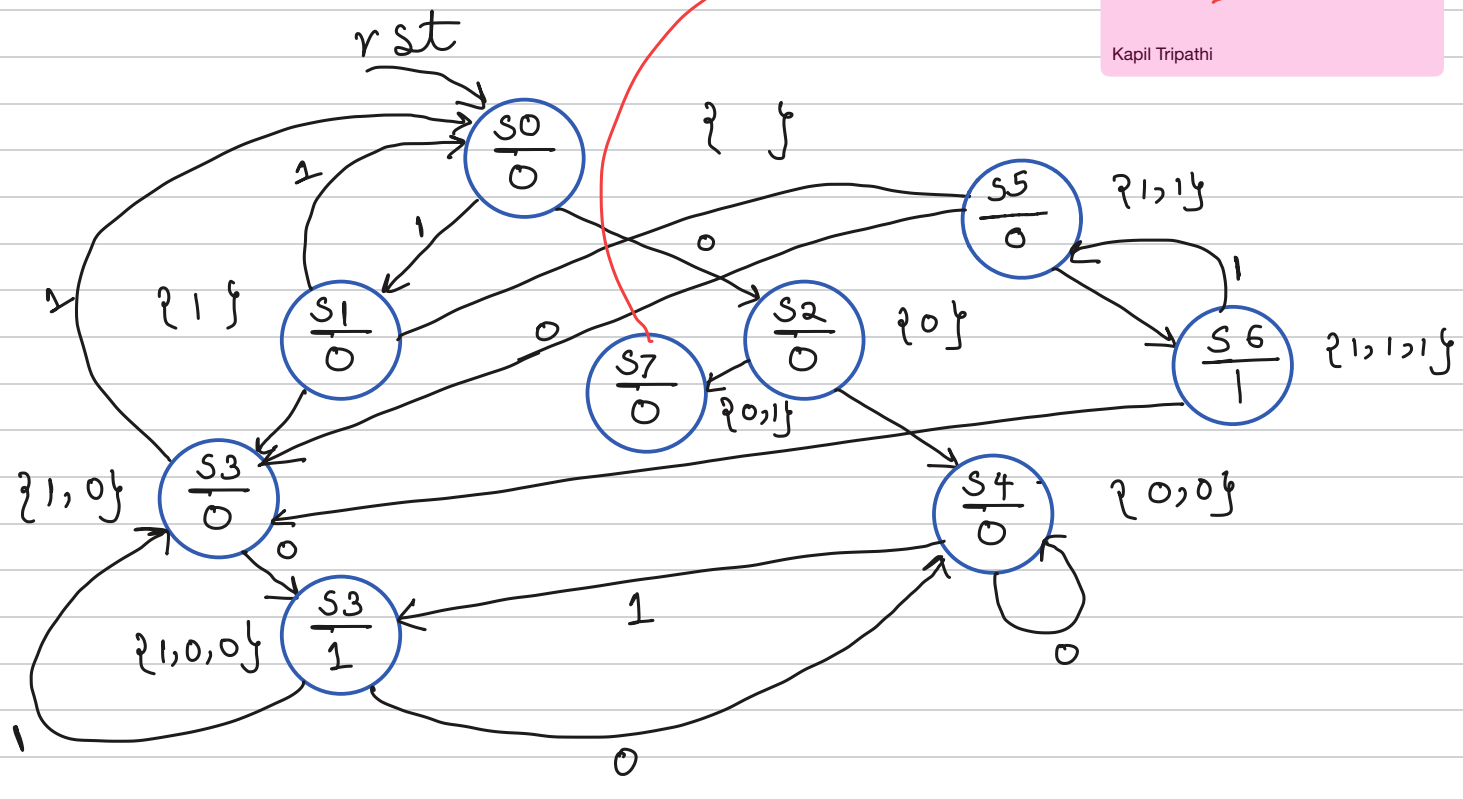
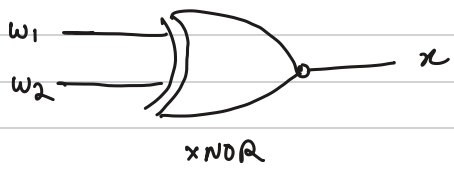
Solution 4.



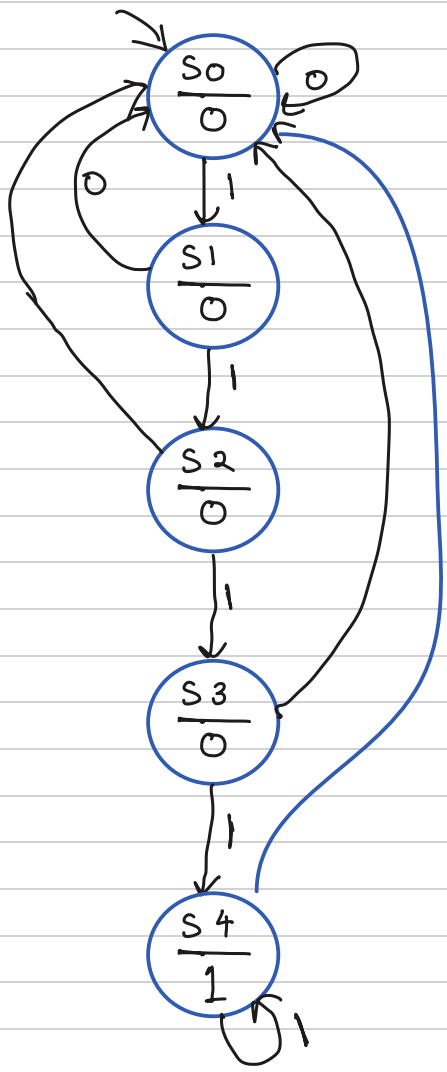
Confusion ?

Kapil Tripathi

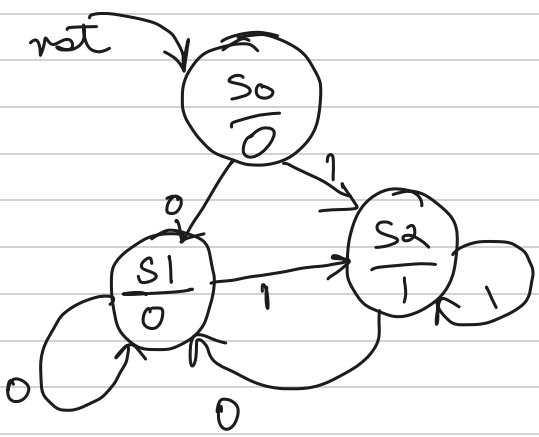
Solution 5.

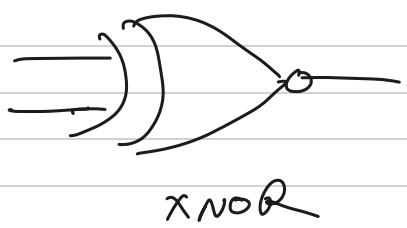


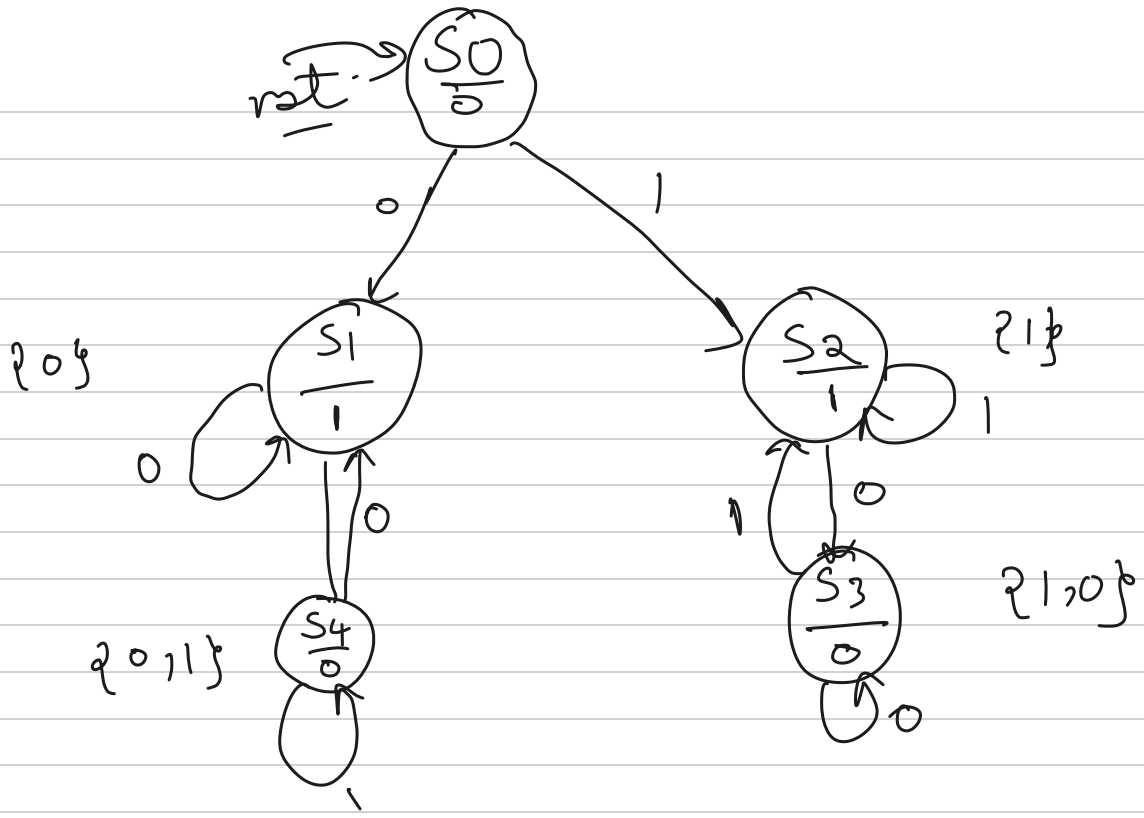
x will be input of the form.



Solution 6. if first bit and current bit are equal





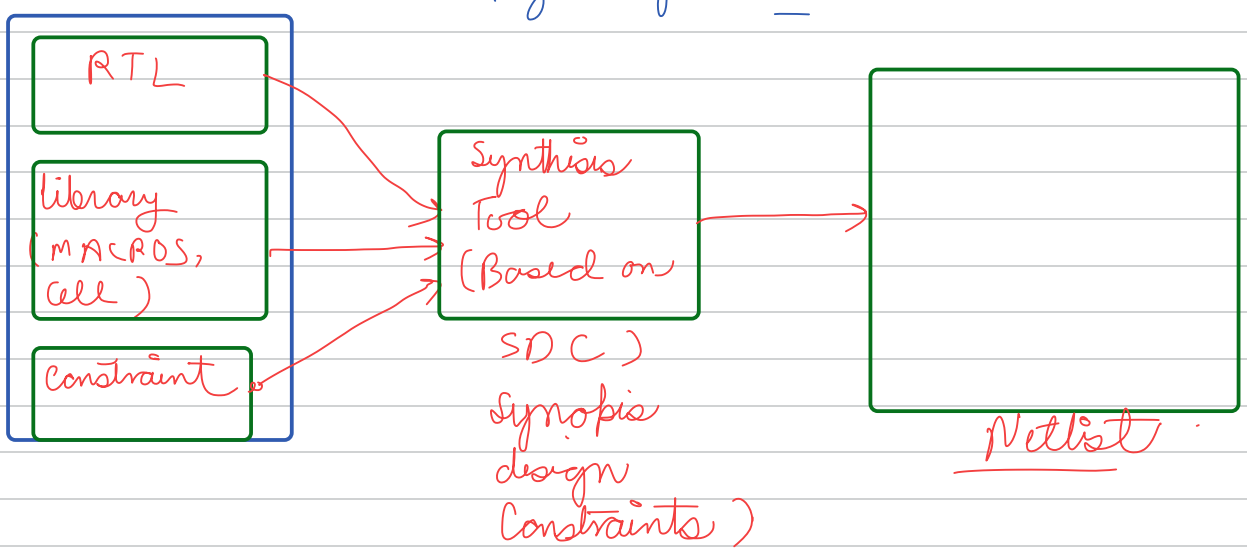


Week 2

behavioural Synthesis — (1) Programs both.

logical synthesis — (2) (RTL synthesis (RTL → generic logic gates)

logic Synthesis.



→ Define, Pin, Port, instance pins, instance, net

I I/A

instance — Pin A

Cell inside a design is instance.

- net wire that connects diff instances & Ports.

Synthes

logical synthesis — (2) b (RTL synthesis (RTL → generic logic gates)

RTL
↓
Netlist (generic logic Gates)
↓
optimized Netlist (logic optimization)
(GenericGate)
↓
Netlist (Standard Cell)
↓
optimized Netlist)

PPA improve
Area driven.
logic optimization (How if Variables ↑)
Technology Mapping (Based on design + Constraints)
We Can Make Delay, Power, timing / More Accurate optimization

Physical design

Logic Synthesis → Netlist

LEF format → library.
Resistance (Bounding box of all & Resistance) Pin location.
liberty

wire + Routing info → Constraints (SDC)

Pad location, shape of die. → Floor Plan

Physical Design Tool.

Placement
Routing.

① Chip Planning
↓
Placement
↓
CTS
↓
Global Routing
↓
Detailed Routing
↓
ECO
↓
Wire GDS.

Floor planning + PDN.

- Chip Planning.
  - Partitioning into subsystems.
  - location of standard cells.
  - Allocation of area for the standard cells, macros, memory.
  - Power Planning (PDN)

② Placement.
→ location of standard cells.
— highly automated.
— wire length minimization (Not have done Routing yet)
→ Timing is met. ; Delay in critical path ↓
→

Diff B/w chip Planning & Placement.

③ CTS.
Lessen clock skew.

④ Routing → Global → Divide into Routing regions.
Local — Allocating wires on each layer.

⑤ ECO

⑥

Verification

→ Simulation.
(Not exhaustive)

— Model checking
formal methods.
CEC - combinational equivalence checking.
↳ functional equivalent using formal methods.

STA → statically Adjacent flip flop.

Physical Design Verification -

DRC

ERC

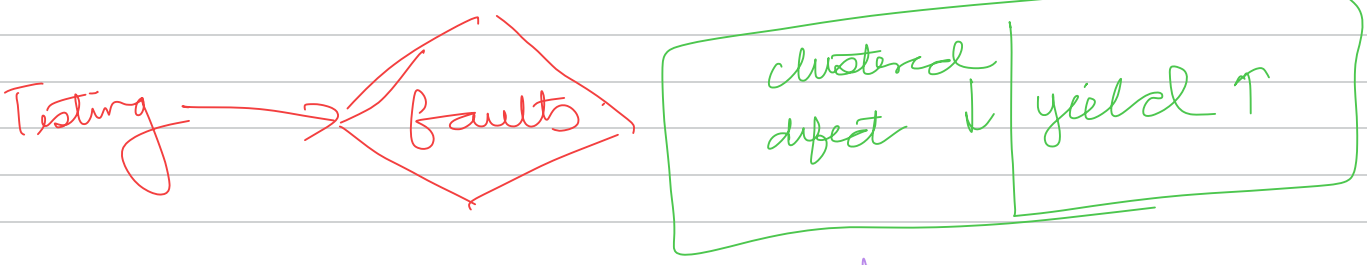
LVS — layout vs schematic Check.

Rules ->

Testing During Design step.

fabrication.

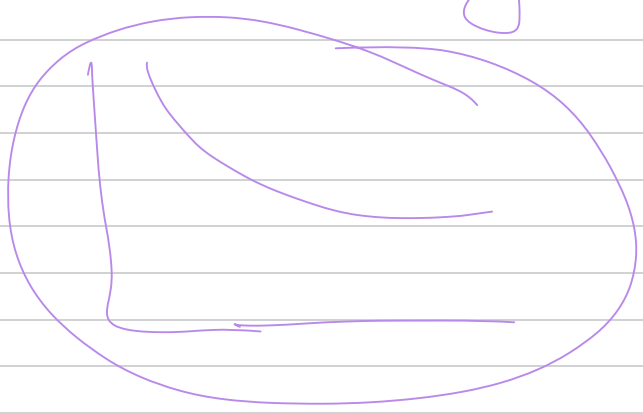
material Property.
furance temperature —
Particles.
—
Deviation in mask features —
S.C defects -



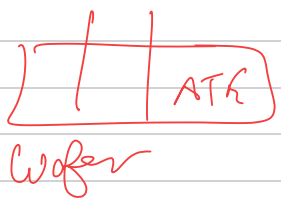
yield Model

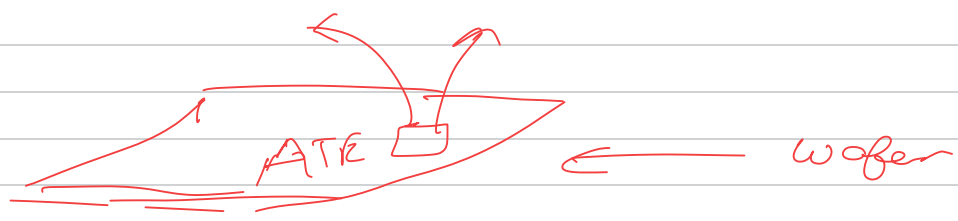
$$y = 1 + \left(\frac{A d}{\alpha}\right)^{-\alpha}$$

A → Area, d → defect density, α → clustering Parameter









Fault Coverage = $\frac{\text{No of faults detectable}}{\text{faults possible}}$.

100

yield = 90

10 chips

Coverage is 50%.

Fault Coverage

10 chip 

$\frac{5}{95} \times 10^6$ 

5 will go To customer

5 will go escape testing.

193 nm Wavelength

Resolution Enhancement -

Add features on the mask. (OPC) optical Proximity Corrections.

→ Double / multiple Patterning - Decompose features in the layout.

FEOL — Caps, diodes etc →

BEOL — Back end of line — Metallic layer.

DIP — Dual in line Package

BGA — Ball grid Array.

Packaging : Binning → on chip delay measurements circuitry :